Decision Tree

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CM22009 – Machine Learning

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Recap: SVM

SVM with Soft Constraints

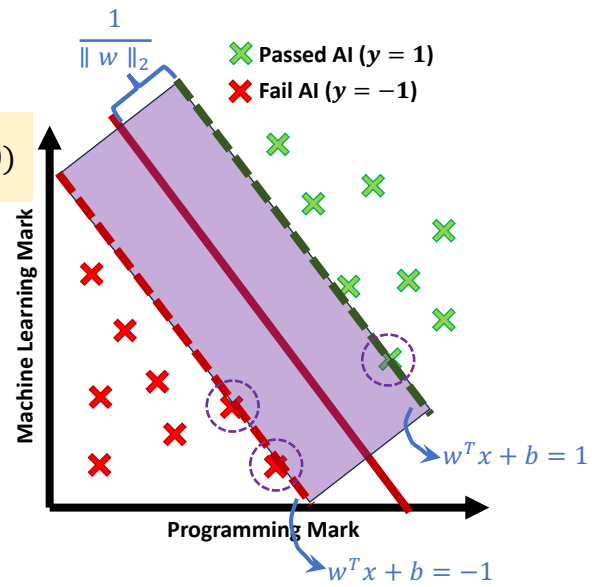
$$\min_{w,b} \frac{1}{2} \| w \|^2 + C \sum_{i=1}^{N} \max(1 - y_i(w^T x_i + b), 0)$$

Kernel Trick: Implicitly map inputs into a higher-dimensional space (maybe infinite) using a kernel function.

$$x \in \mathbb{R}^D$$

$$K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$$

(one commonly used kernel)



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Where are we?

Week No.	Lecture A	Lecture B
1 (29/09/2025)	Intro. Supervised Learning	Logistic Regression
2 (06/10/2025)	Nearest-Neighbours	Support Vector Machines
3 (13/10/2025)	Decision Trees	Random Forests
4 (20/10/2025)	Naïve-Bayes	Data Engineering
5 (27/10/2025)		
6 (03/11/2025)	Reading Week	

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Learning Objectives

Describe the components and purpose of decision trees.

Build decision trees using entropy and information gain.

Explain how to handle numerical features to build decision trees.

Discuss the strengths and weaknesses of decision trees and explain strategies to address their key issues.

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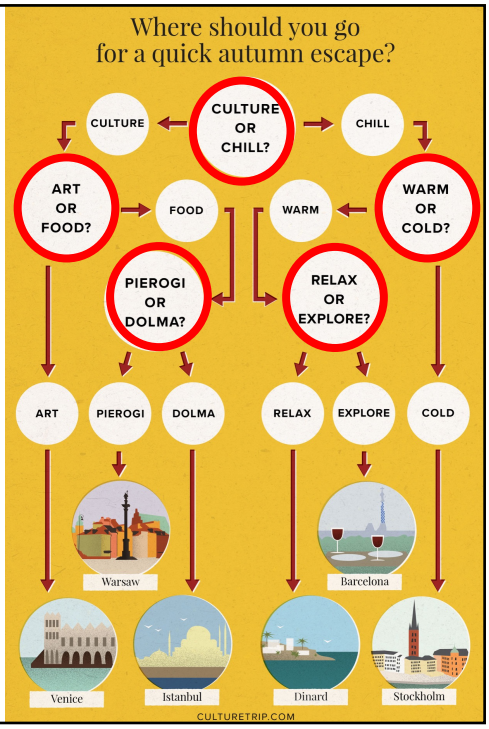
Decision Tree: A supervised learning model that predicts a target variable by recursively splitting data into a tree-like structure of decisions.

- Mimics human decision-making using a tree-like structure.
- Main components:
  - Root node
  - Decision nodes: represent choices to be made
    - Yes or no question based on feature values
    - Multiple choices



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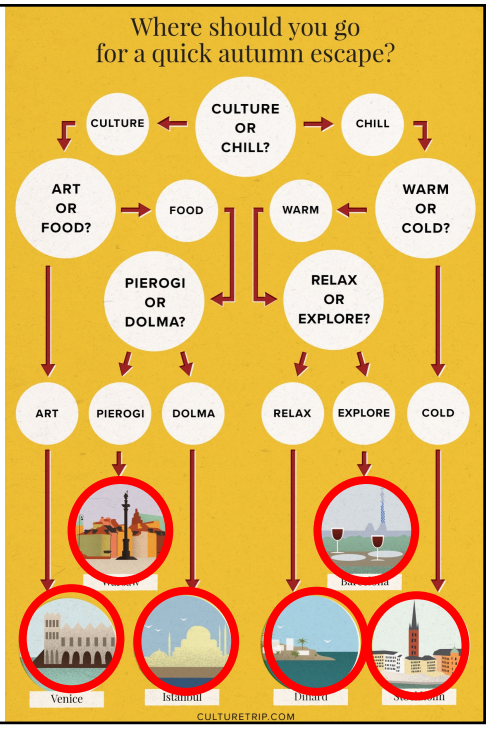
Decision Tree: A supervised learning model that predicts a target variable by recursively splitting data into a tree-like structure of decisions.

- Mimics human decision-making using a tree-like structure.
- Main components:
  - Root node
  - Decision nodes: represent choices to be made
    - Yes or no question based on feature values
    - Multiple choices
  - Leaf nodes: final outcomes (class or value)
- Used for both classification and regression tasks



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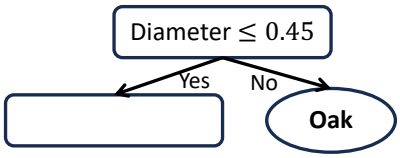
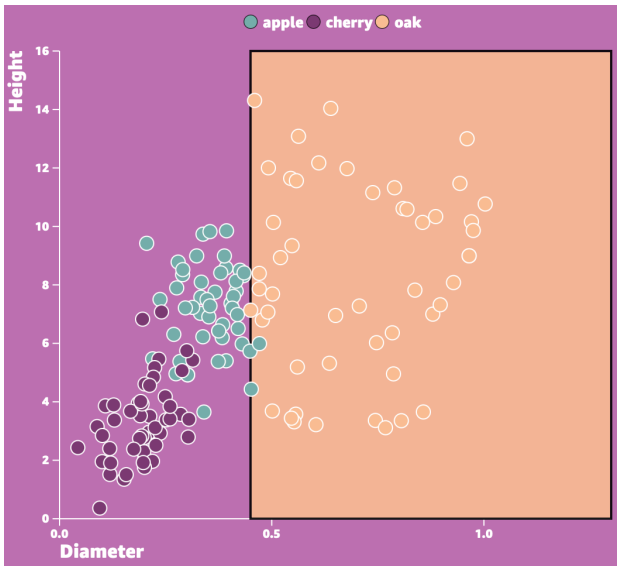
One example





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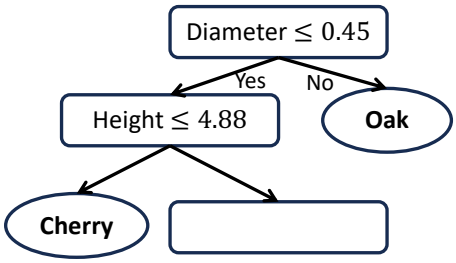
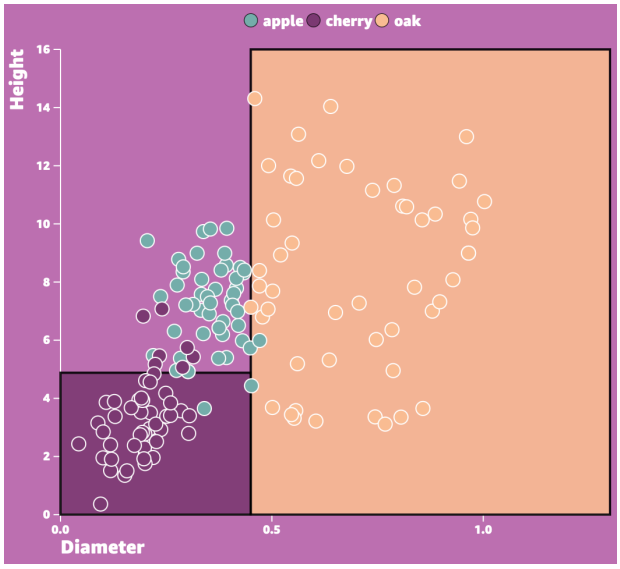
One example





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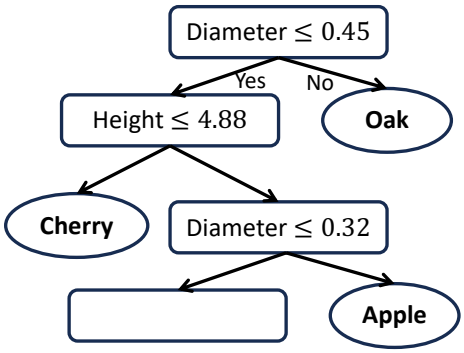
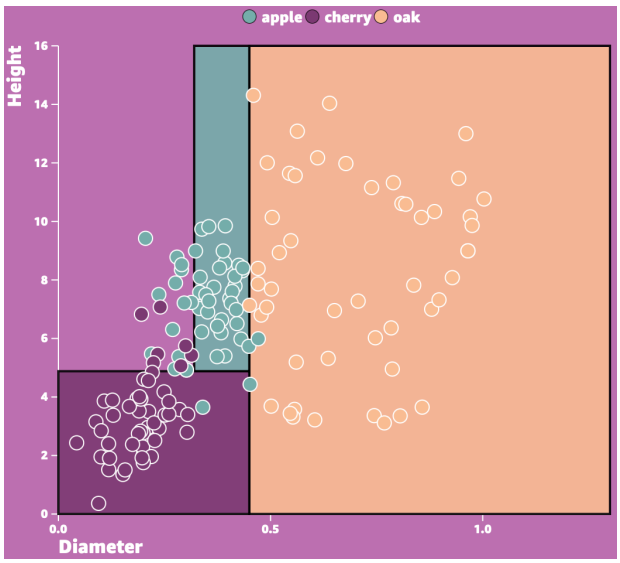
One example





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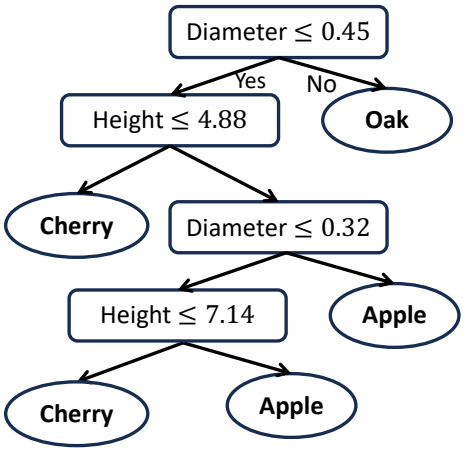

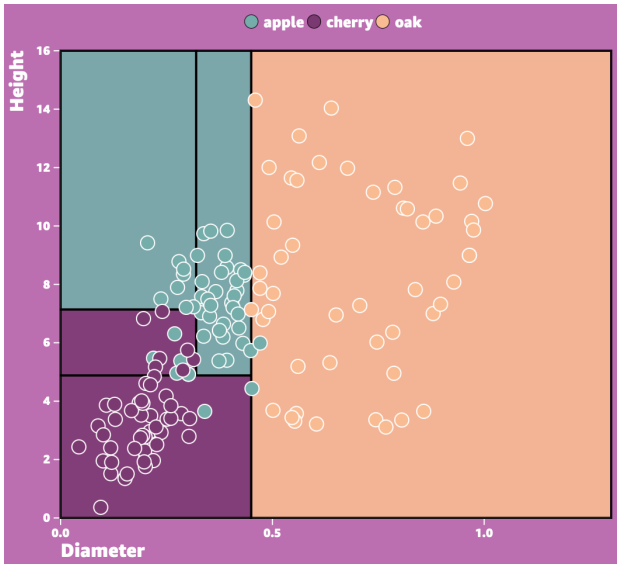

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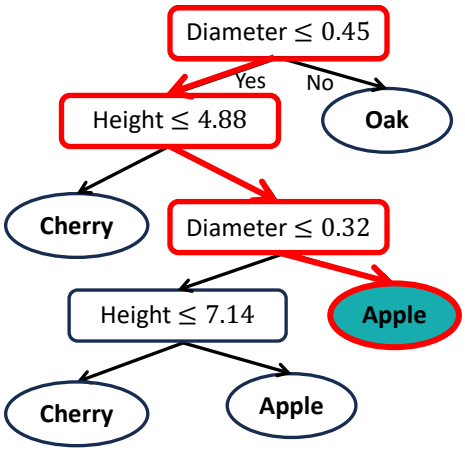
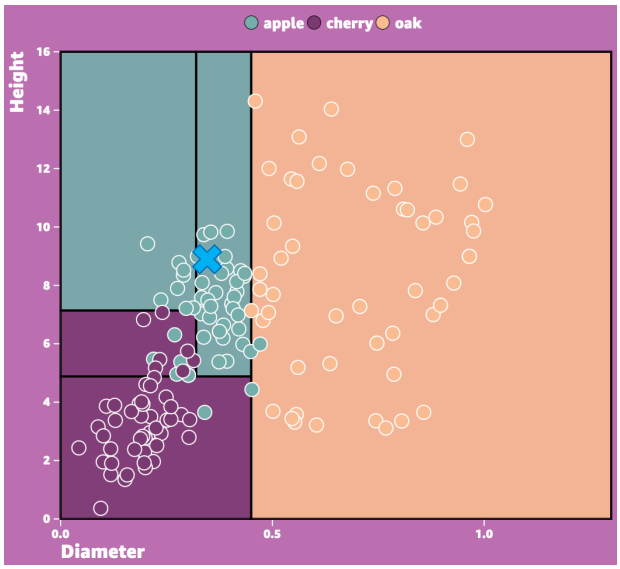
One example



Q: What tree is this (Diameter=0.35, Height=8.9)?



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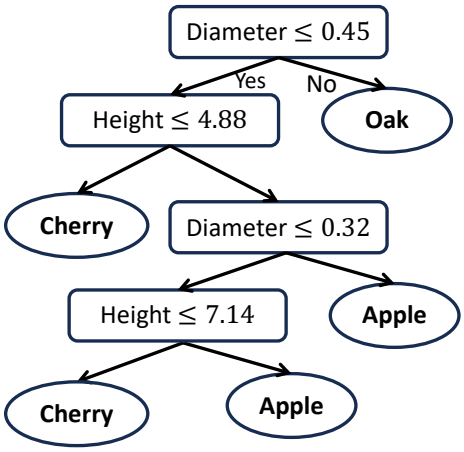
One example



Each internal/decision node choose a feature and threshold that best separates the data

Q: How to best split the data?

Img from: https://mlu-explain.github.io/decision-tree/?utm_source=chatgpt.com

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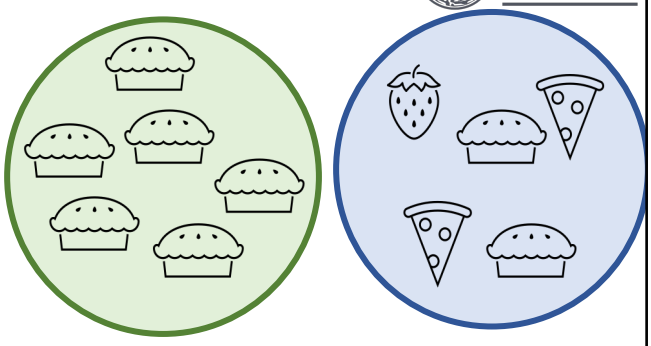
Entropy

- In thermodynamics: entropy measures **disorder or randomness** in a physics system.
- In information theory: entropy measures the **uncertainty** of the sample values in information.



The outcome is very predictable

The outcome is very uncertain

Low entropy (near 0)

High entropy (near 1)

Img from: https://mlu-explain.github.io/decision-tree/?utm_source=chatgpt.com

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Entropy

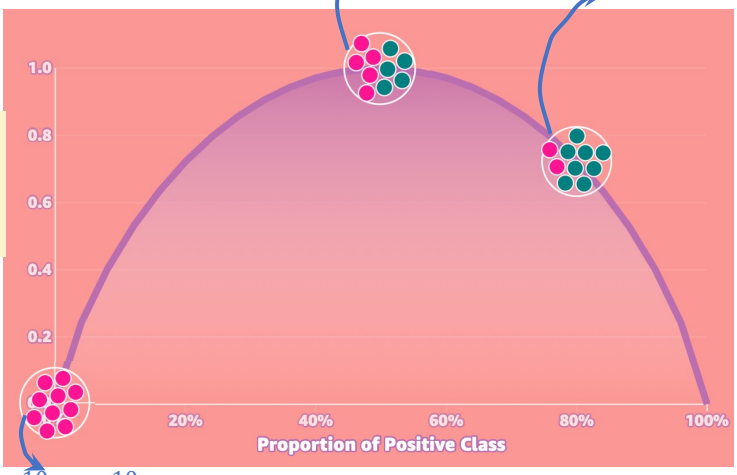
$$-\left(\frac{5}{10}\log_2\left(\frac{5}{10}\right)+\frac{5}{10}\log_2\left(\frac{5}{10}\right)\right)=1$$

$$-\left(\frac{2}{10}\log_2\left(\frac{2}{10}\right)+\frac{8}{10}\log_2\left(\frac{8}{10}\right)\right)=0.722$$

$$E(S)=-\sum_{c=1}^{C}\boxed{p_c}\log_2(p_c)$$

Probability of class c

The fraction of datapoints of class c in the decision node S



$$-\left(\frac{0}{10}\log_2\left(\frac{0}{10}\right)+\frac{10}{10}\log_2\left(\frac{10}{10}\right)\right)=0$$

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Information gain: how much uncertainty decrease after the split.

Entropy before splitting

$\mathcal{A}$: all possible values for feature a

For categorical data: $$IG(S,a) = E(S) - \underbrace{\sum_{v \in \mathcal{A}} \frac{|S_v|}{|S|} E(S_v)}_{\text{Entropy after splitting with feature } a \text{ (weighted average entropy on all child nodes)}}$$

a: A specific attribute/feature

Q: When we split at a decision node, do we aim to decrease or increase the entropy?

A: Decrease the entropy.

Aim for splitting a node S : $\max_a IG(S,a)$

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Example: Will a customer buy one laptop?

- **STEP 1**: Calculate the initial entropy

Yes: 8, No: 4

$$E(S) = -\sum_{c=1}^{2} p_c \log_2(p_c)$$
$$= -\frac{8}{12}\log_2\left(\frac{8}{12}\right) - \frac{4}{12}\log_2\left(\frac{4}{12}\right)$$
$$= 0.918$$

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

STEP 2.1 Feature 1 (Age)

Young: Yes: 2, No: 3

$$E(\text{Age_young}) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)$$

$$= 0.971$$

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

STEP 2.1 Feature 1 (Age)

Young: Yes: 2, No: 3

Middle: Yes: 3, No: 0

$$E(\text{Age_middle}) = -\frac{3}{3}\log_2\left(\frac{3}{3}\right) - \frac{0}{3}\log_2\left(\frac{0}{3}\right)$$

$$= 0$$

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

STEP 2.1 Feature 1 (Age)

Young: Yes: 2, No: 3

Middle: Yes: 3, No: 0

Senior: Yes: 3, No: 1

$$E(\text{Age_senior}) = -\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)$$
$$= 0.811$$

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

STEP 2.1 Feature 1 (Age)

Young: Yes: 2, No: 3

Middle: Yes: 3, No: 0

Senior: Yes: 3, No: 1

$$E(S|\text{Age}) = \frac{5}{12}E(\text{Age_young}) + \frac{3}{12}E(\text{Age_Middle}) + \frac{4}{12}E(\text{Age_senior})$$
$$= 0.675$$

Entropy after splitting with feature 'Age'

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

STEP 2.2 Feature 2 (Income)

High: Yes: 1, No: 2

Medium: Yes: 4, No: 1

Low: Yes: 3, No: 1

$$E(S|\text{Income}) = \frac{3}{12}E(\text{Income_High}) + \frac{5}{12}E(\text{Income_Medium}) + \frac{4}{12}E(\text{Income_Low}) = 0.801$$

Entropy after splitting with feature 'Income'

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

STEP 2.3 Feature 3 (Student)

No: Yes: 3, No: 3

Yes: Yes: 5, No: 1

$$E(S|\text{Student}) = \frac{6}{12}E(\text{student_No}) + \frac{6}{12}E(\text{Student_Yes}) = 0.825$$

Entropy after splitting with feature 'Student'

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

✓ $IG(S, \text{Age}) = E(S) - E(S|\text{Age}) = 0.918 - 0.675 = 0.243$

$IG(S, \text{Income}) = E(S) - E(S|\text{Income}) = 0.918 - 0.801 = 0.117$

$IG(S, \text{Student}) = E(S) - E(S|\text{Student}) = 0.918 - 0.825 = 0.093$

- **STEP 3**: Choose the best split (with largest information gain).

Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
Young	Medium	No	No
Young	Low	Yes	Yes
Senior	Medium	Yes	Yes
Young	Medium	Yes	Yes
Middle	Medium	No	Yes

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Example: Will a customer buy one laptop?

- **STEP 2**: Calculate information gain for each feature

✓ $IG(S, \text{Age}) = E(S) - E(S|\text{Age}) = 0.918 - 0.675 = 0.243$

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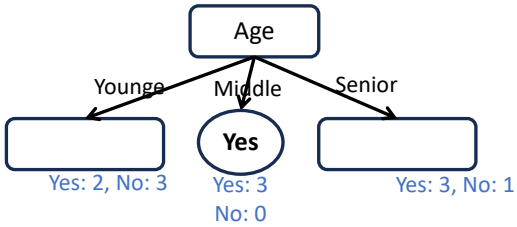
$IG(S, \text{Student}) = E(S) - E(S|\text{Student}) = 0.918 - 0.825 = 0.093$

- **STEP 3**: Choose the best split (with largest information gain).



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Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
Middle	High	No	Yes
Senior	Medium	No	Yes
Senior	Low	Yes	Yes
Senior	Low	Yes	No
Middle	Low	Yes	Yes
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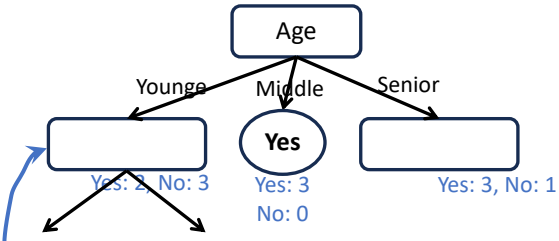

Q: What feature should we choose to split?

- **STEP 1**: Calculate the initial entropy
  Yes: 2, No: 3
- **STEP 2**: Calculate information gain for each feature
- **STEP 3**: Choose the best split (with largest information gain).

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Age	Income	Student	Buys
Young	High	No	No
Young	High	No	No
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Senior	Medium	No	Yes
Senior	Low	Yes	Yes
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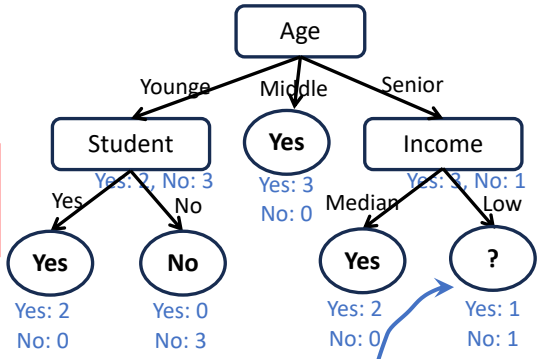

Not possible to split here. Either output the parent's majority class or a random class.

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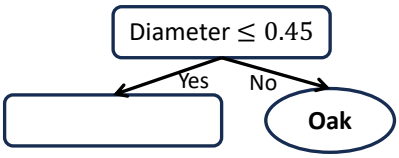
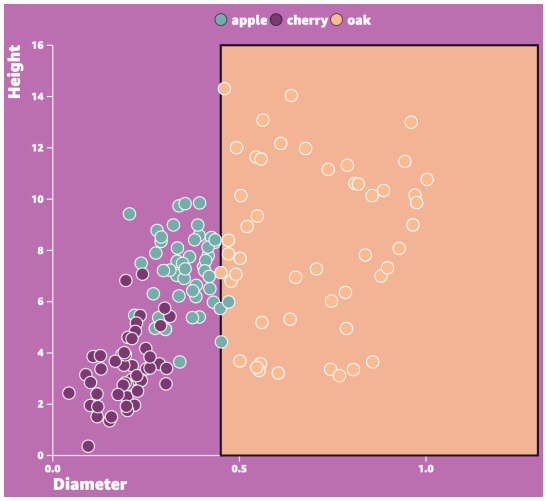
How to handle continuous data?





Img from: https://mlu-explain.github.io/decision-tree/?utm_source=chatgpt.com

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How to handle continuous data?

For each feature $x_j, j = 1, \dots, D$

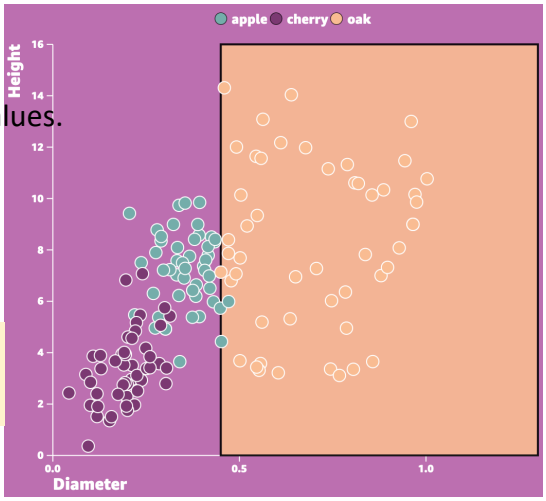
1. Rank order all values of this feature
2. Calculate the midpoints θ between two adjacent values.
3. Calculate the information gain for all midpoints θ.

Choose the j and θ with the highest information gain.

Aim for splitting a node S : $\max_{j,\theta} IG(x_j, \theta)$

$$IG(x_j, \theta) = E(S) - \left(\frac{|L|}{|S|}E(L) + \frac{|R|}{|S|}E(R)\right)$$

L: left node
R: right node



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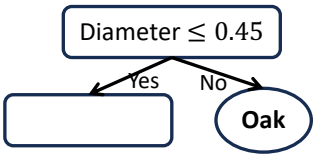
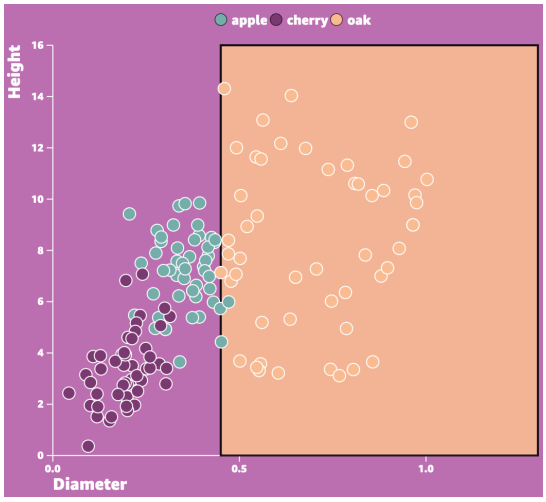
How to handle continuous data?

Recall our first decision node split on **Diameter** $\leq \mathbf{0.45}$, why?





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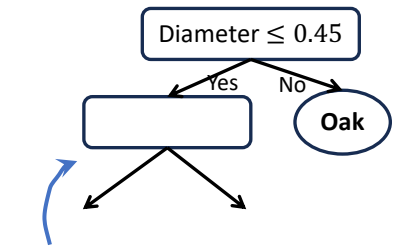
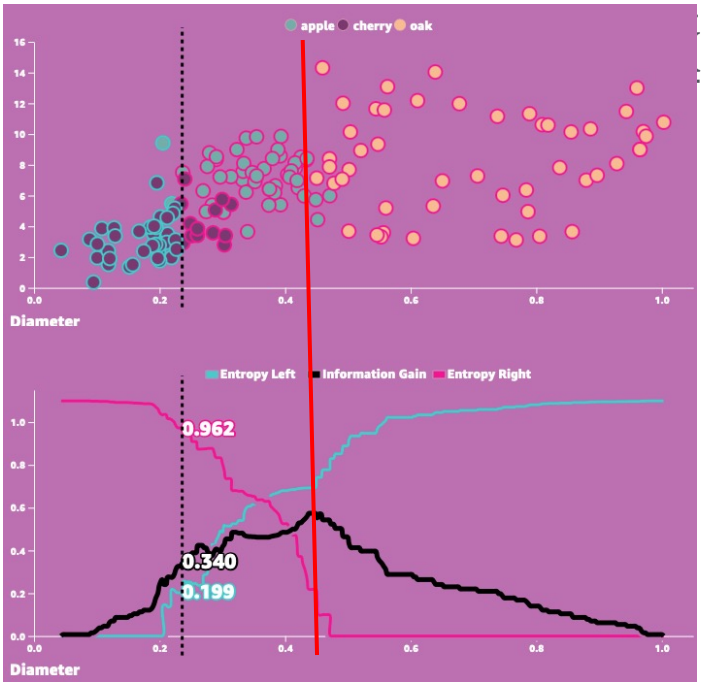
Recall our first decision node split on **Diameter** $\leq \mathbf{0.45}$, why?



Continue to split this node by $\max_{j,\theta} IG(x_j, \theta)$



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Advantages	Disadvantages
Easy to interpret	Prone to overfitting (keep splitting until leaf is pure)
Minimal data preparation (No scaling needed, handle missing values)	Unstable (small change in data → large tree differences)
Versatility (regression, multi-class, categorical & continuous features, non-linear)	Computationally expensive for large datasets

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How to address these issues

- Pre-pruning (early stopping)
  - Limit depth, minimum samples per leaf, or minimum information gain
- Post-pruning
  - Grow full tree first, then prune branches that don't improve accuracy (e.g., using cross validation)
- Ensemble methods (most effective)
  - Combine many trees (e.g., Random Forest, Gradient Boosting) – average or weighted voting reduces overfitting

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Summary

- Decision tress learn decision rules in a simple, interpretable way
- Split criteria: choose a split with largest **information gain**

For categorical data: $\max_{a} IG(S, a)$

$$IG(S, a) = E(S) - \sum_{v \in \mathcal{A}} \frac{|S_v|}{|S|} E(S_v)$$

For continuous data: $\max_{j,\theta} IG(x_j, \theta)$

$$IG(x_j, \theta) = E(S) - \left(\frac{|L|}{|S|} E(L) + \frac{|R|}{|S|} E(R)\right)$$

- Alternative split criteria: **Gini impurity** (measure how "impure" a node is).

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Feedback!

If you have any feedback, use the QR code or link to anonymously report it.

I'm open to any comments/feedback.

SCAN ME

https://forms.gle/AyTC145yPeFVMiQw8

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